



FIXING THE GAPS - (Executive Governance Panel)

Pakistan Education Performance | Government Education Governance

Year

All

Province, City

All



Clear Filters

Total Schools



608K

Primary Schools



489K

Secondary Schools



120K

Education Score



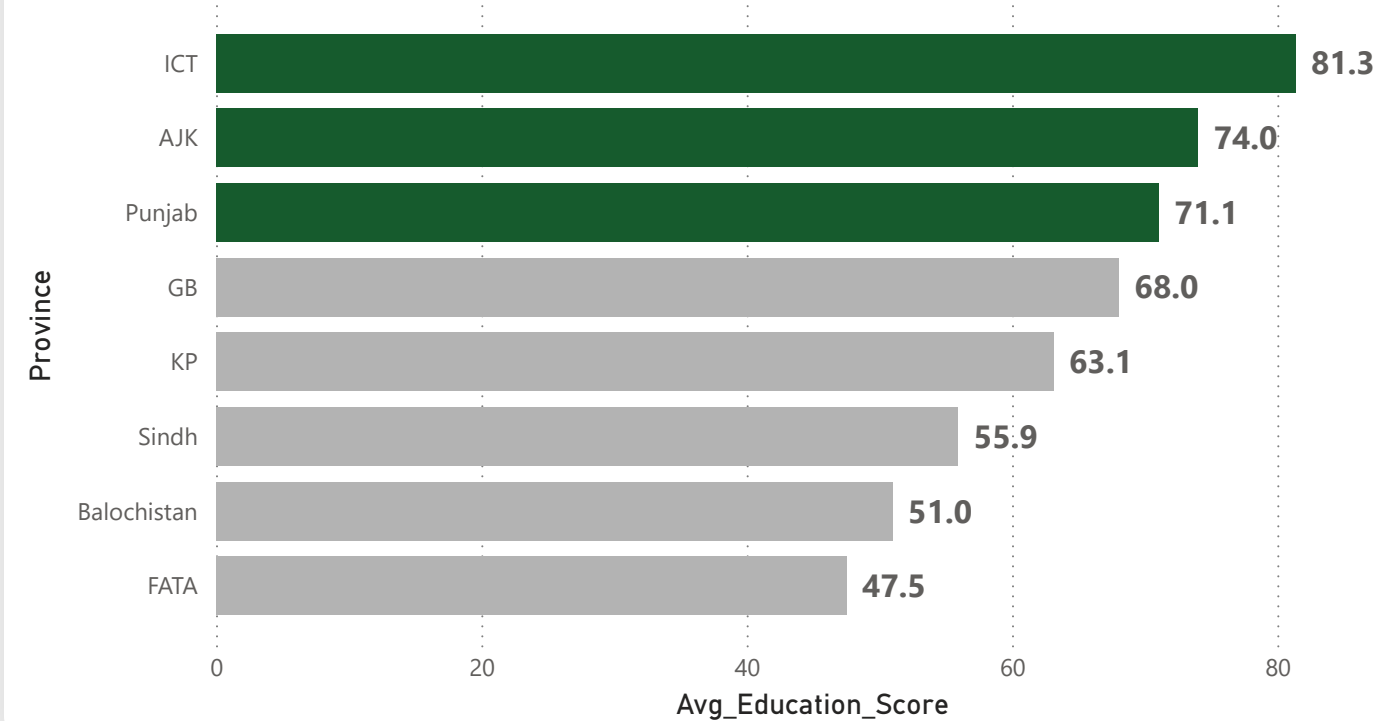
61.15

Critical Gender Gap Regions



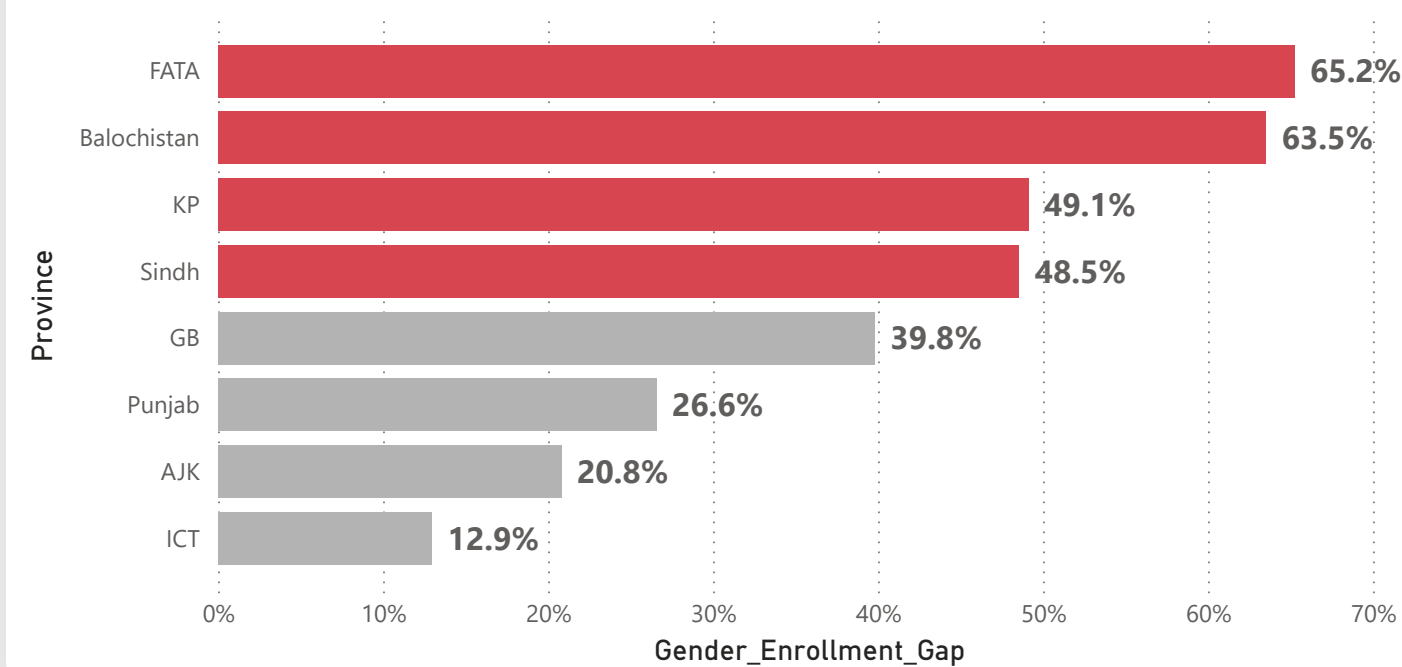
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Average Education Score by Province

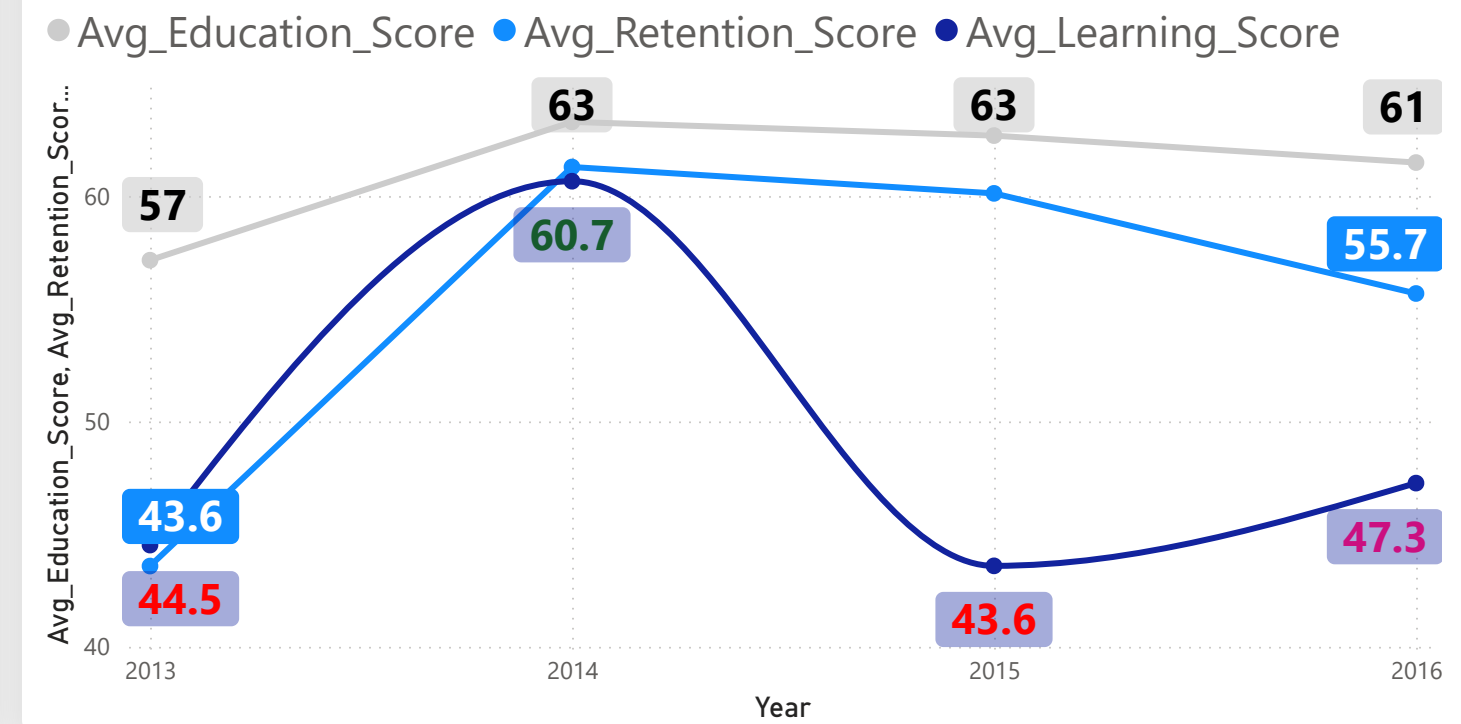


Gender Enrollment Gap by Province

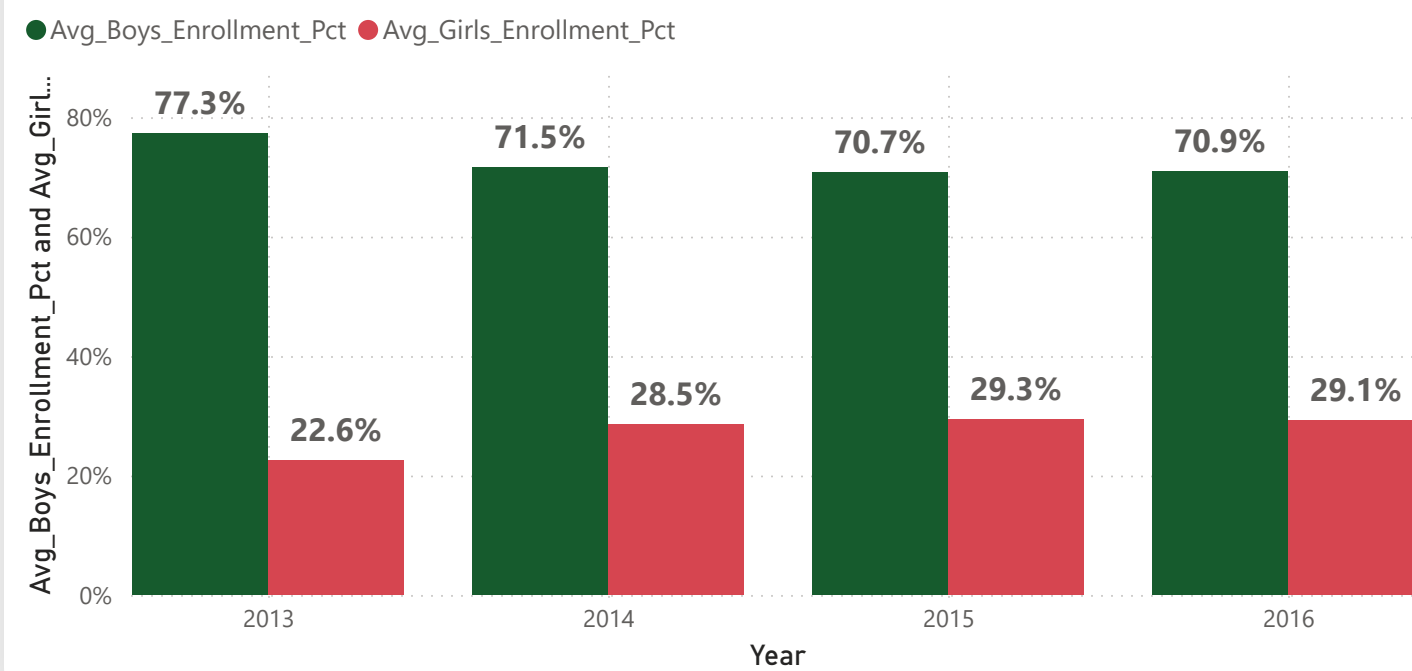
How does Pakistan's education situation look?



Education, Learning & Retention Trends



Boys vs Girls Enrollment Trend



Regional Infrastructure Policy Context

Comparing essential facility coverage and infrastructure gaps across provinces and districts

Province	Total_Schools	Total_Primary_Schools	Total_Secondary_Schools	Avg_Education_Score	Gender_Enrollment_Gap	Gender_Gap_Risk
ICT	1564	764	800	81.34	12.95%	Low
AJK	23940	16808	7132	74.00	20.79%	Low
Punjab	211944	151412	60532	71.05	26.55%	Low
GB	5100	3188	1912	68.04	39.79%	Low
KP	112712	92088	20624	63.11	49.11%	Moderate
Sindh	177584	160544	17040	55.93	48.53%	Moderate
Balochistan	51468	43240	8228	50.98	63.50%	Moderate
FATA	24044	20456	3588	47.50	65.23%	Moderate

Source: Pakistan Education Performance Dataset | Kaggle | Historical district-level data, 2013–2016



Executive Governance

Risk & Intervention

Infrastructure Audit

Infrastructure Analysis

Policy Impact & Socio...

Socio Economic...

FIXING THE GAPS - Executive Risk & Intervention

Anomaly Detection | Priority Areas | Recommended Actions

Year

All

Region

All

Risk View

Overall View

Clear Filters

Critical Gender Gap Regions



Critical Infrastructure Regions



Priority Intervention Regions

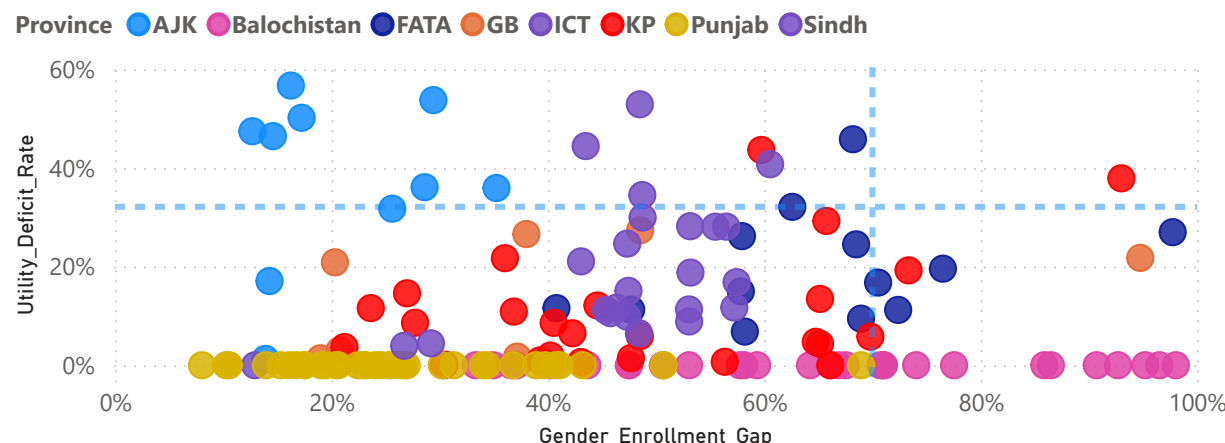


Overall Risk Score



Gender Gap vs Infrastructure Deficit

Identifying regions requiring combined intervention

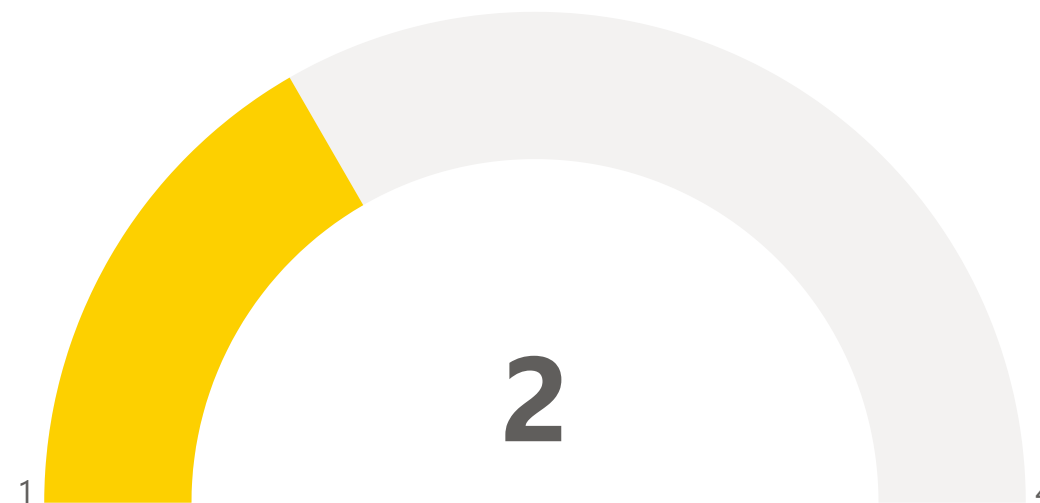


KEY INSIGHTS

KP (Kohistan) shows very high gender gap and infrastructure deficit, indicating urgent need for combined intervention.

Action: Prioritize low performing regions in the top-right quadrant.

Overall Risk



KEY INSIGHTS

KP (Kohistan) risk score is 4 (High Risk) indicating critical challenges in gender equality and infrastructure requiring immediate government action.

Action: Investigate Kohistan's post-2014 decline and review policy

Priority Intervention Matrix

Regions with High/Critical gender and infrastructure risks

Province	Gender_Enrollment_Gap	Avg_Retention_Score	Avg_Learning_Score	Gender_Gap_Risk	Infrastructure_Risk
AJK	20.8%	71.1	60.5	Low	High
Balochistan	63.5%	43.0	38.3	Moderate	Low
FATA	65.2%	38.8	37.3	Moderate	Moderate
GB	39.8%	68.3	60.1	Low	Moderate
ICT	12.0%	81.2	64.6	Low	Low

KEY INSIGHTS

Kohistan has critical gender gap (93%) and high infrastructure deficit (37.9%) with high teacher and classroom strain.

Priority Region — Kohistan

45.18%

Gender_Enrollment_G...

10.26%

Utility_Deficit_Rate

49.0

Avg_Learning_Score

55.2

Avg_Retention_Score

KEY INSIGHTS

Kohistan shows critical gender gap and high infrastructure deficit, resulting in low learning and retention scores.



Recommended Government Action

No immediate priority intervention identified.

✓ Improve Basic Facilities

Ensure availability of electricity, drinking water, toilets and classrooms in al primary schools.

✓ Increase Girl's Enrolment

Implement awareness and community engagement programs to promote girl's education.

✓ Reduce Teacher & Classroom Strain

Improve teacher deployment and build additional classrooms to reduce overcrowding.

KEY INSIGHTS

Targeted infrastructure improvements and girl's education initiatives can substantially improve learning and retention outcomes in high-risk regions like Kohistan.



DISTRICT INFRASTRUCTURE AUDIT

Infrastructure, Facilities & Staffing Risk

Year

All

Province

All

Clear Filters

Utility Deficit Rate



10.26%

Average Drinking Water



59.68

Single-Teacher Strain



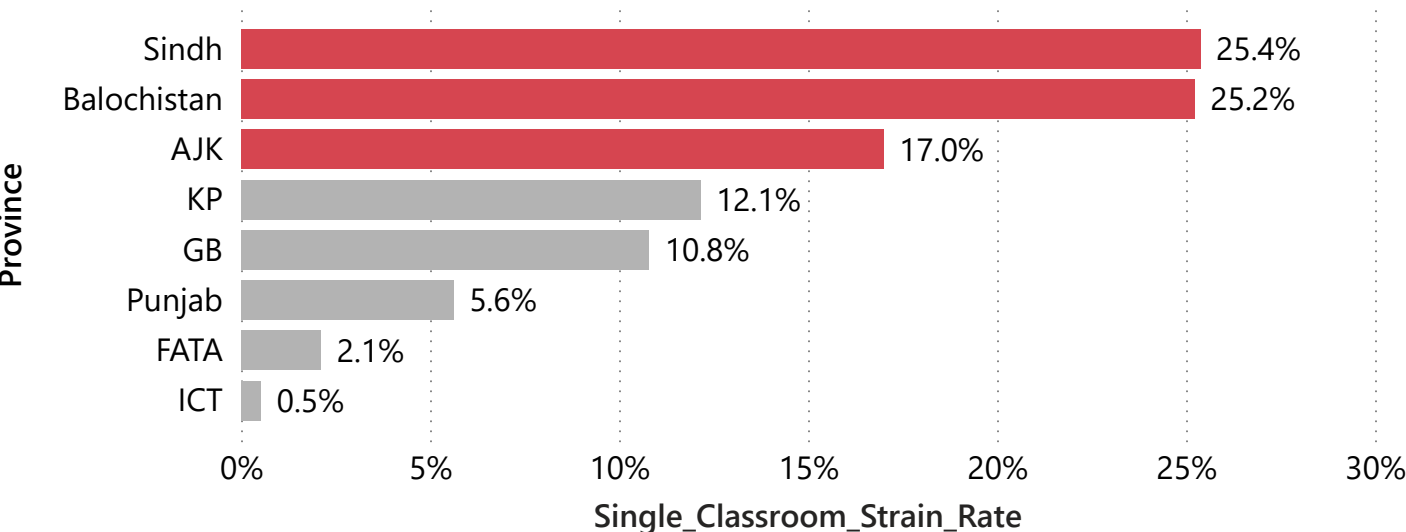
29.0%

Single-Classroom Strain

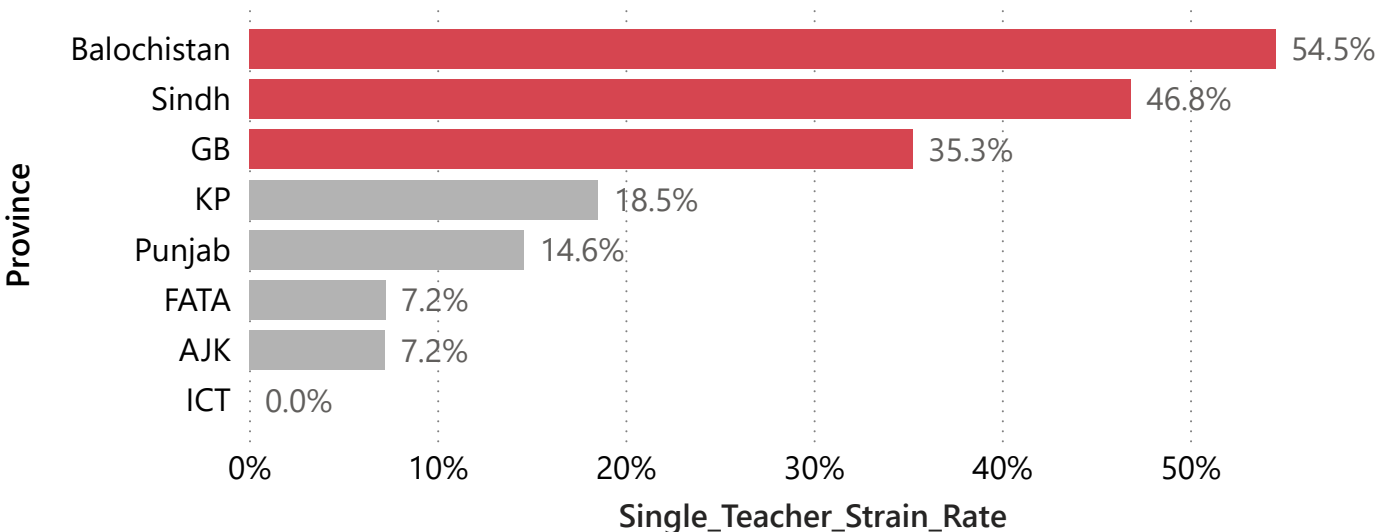


15.3%

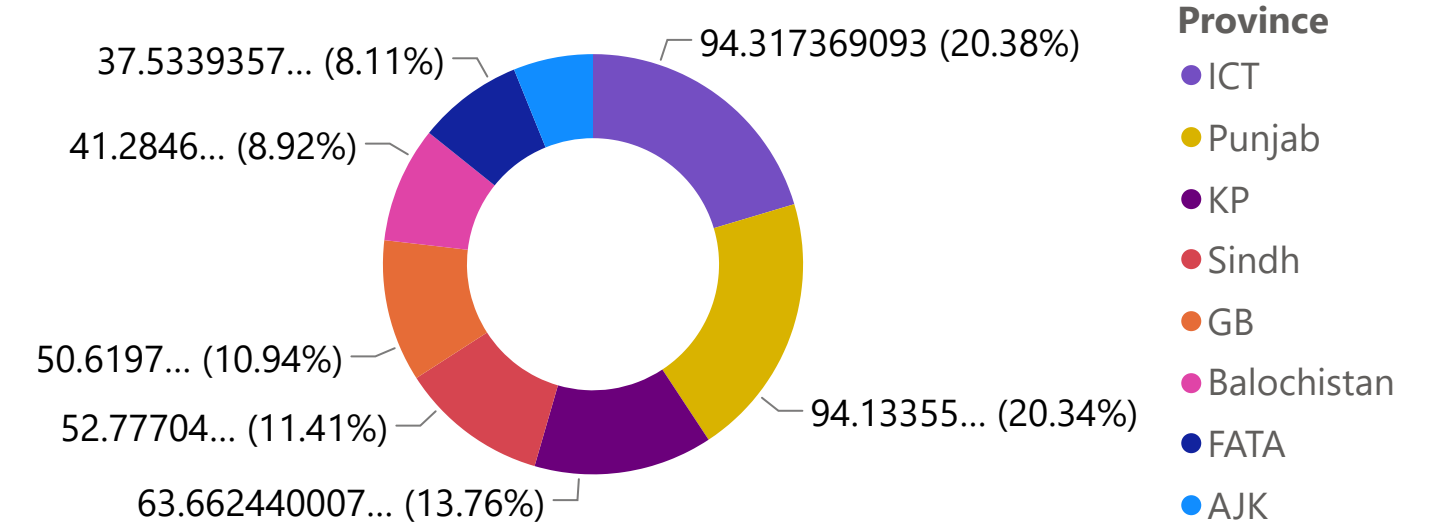
Single-Classroom Strain by Province



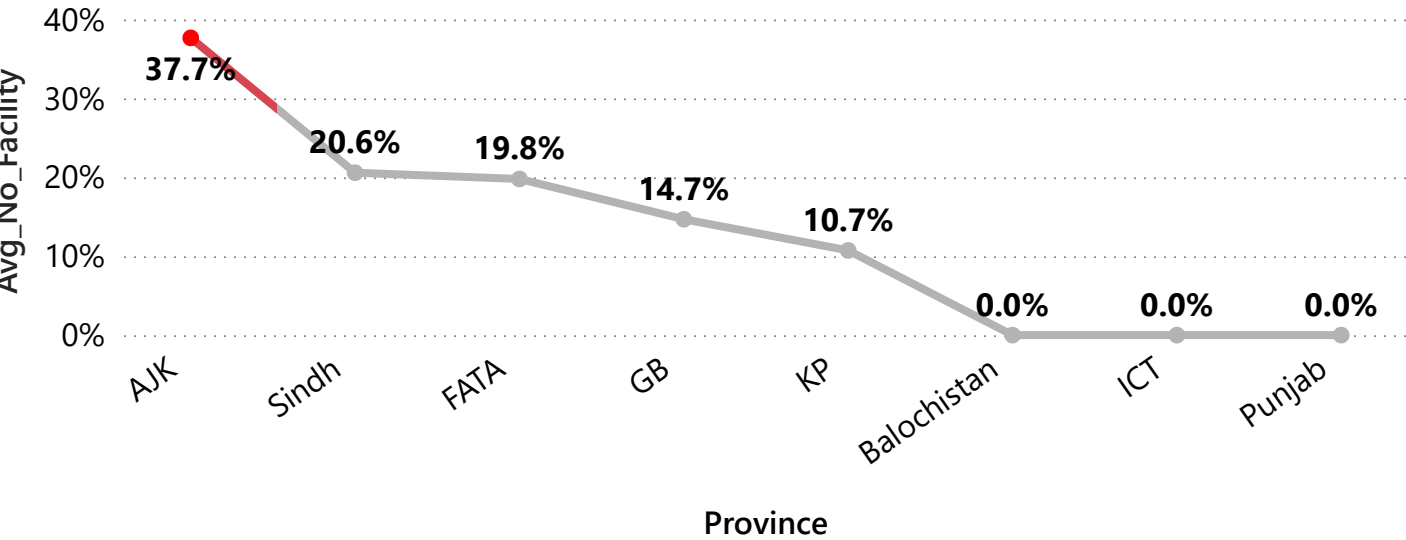
Single-Teacher Strain by Province



Avg_Drinking_Water by Province



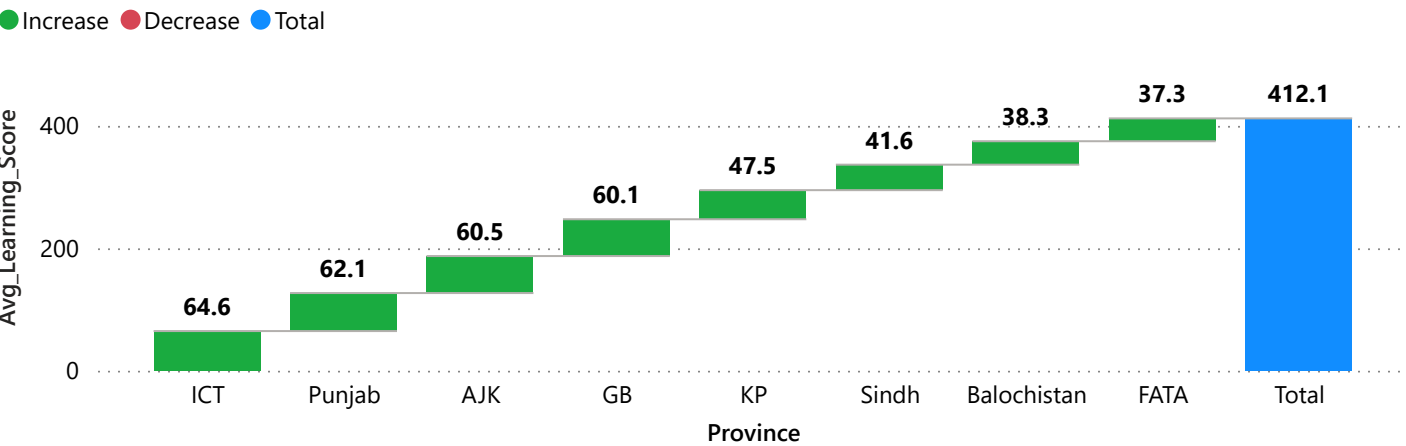
Average No Facility by Province



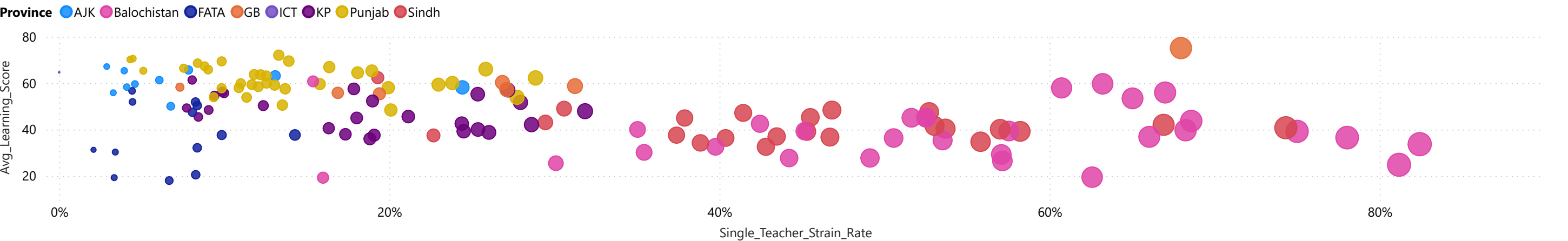
District Infrastructure Audit

Province	Single_Classroom_Strain_Rate	Avg_Electricity	Avg_Drinking_Water	Avg_Toilet	Avg_Building_Condition	Infrastructure_Risk
AJK	17.0%	14.92	28.38	30.08	34.63	High
Balochistan	25.2%	26.55	41.28	22.34	21.05	Low
ICT	0.5%	98.33	94.32	96.83	46.21	Low
KP	12.1%	53.16	63.66	76.24	70.75	Low
Punjab	5.6%	75.35	94.13	87.66	77.17	Low
FATA	2.1%	40.48	37.53	37.27	26.44	Moderate
GB	10.8%	43.93	50.62	47.87	24.10	Moderate
Sindh	25.4%	44.10	52.78	50.04	27.11	Moderate

Learning Score by Province



Single-Teacher Strain vs Learning Score





DISTRICT INFRASTRUCTURE ANALYSIS

Infrastructure Deficits • Utility Gaps • District Priorities

Critical Infrastructure Regions

15

Utility Deficit Rate

10.26%

Schools With No Facilities

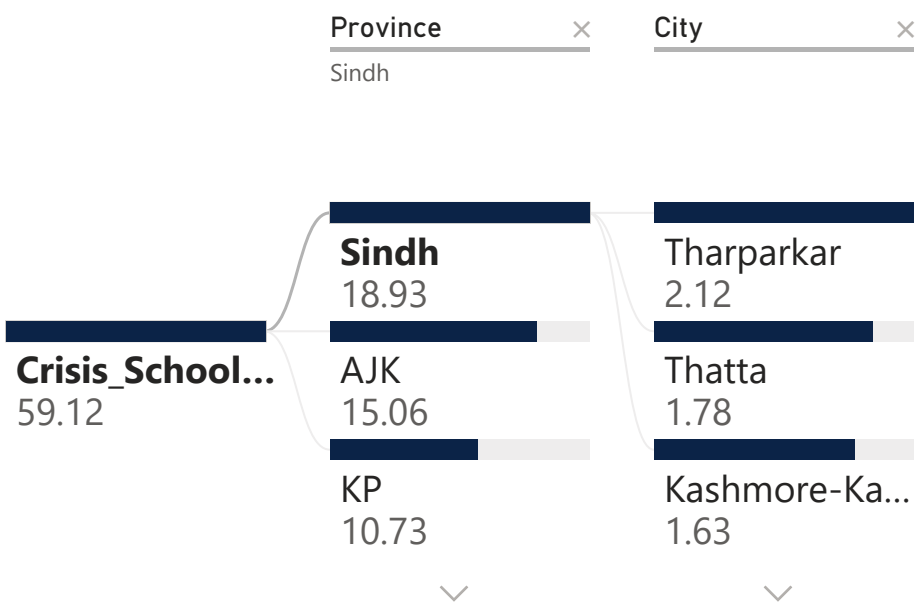
10.26%

Affected-School Districts

8

Q1. Severe Infrastructure Deficits District

Districts with the highest infrastructure deficit rates



KEY INSIGHTS

Several districts in Sindh show severe infrastructure deficit rates above 15%. Tharparkar, AJK and KP regions require immediate infrastructure investment.

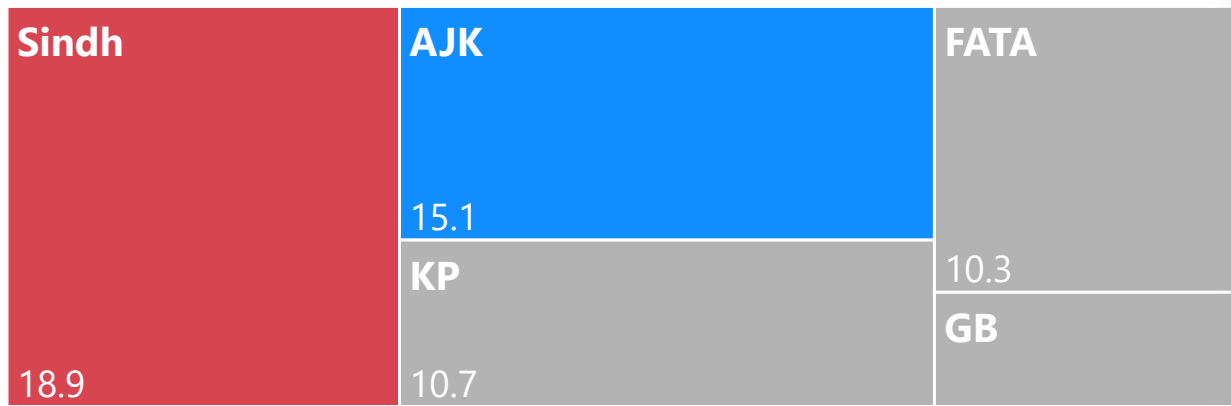
Q2. Basic Facility Coverage & Deficiency by District

Higher coverage is better; higher No Facility indicates greater deficiency

Province	Avg_Electricity	Avg_Drinking_Water	Avg_Toilet	Avg_No_Facility
AJK				
Haveli	3.27	24.83	21.93	56.69%
Hattian	8.60	30.47	42.16	53.74%
Sudhnutti	6.50	12.18	13.83	50.16%
Poonch	5.77	14.43	22.77	47.43%
Kotli	16.61	20.89	18.71	46.44%
Muzaffarabad	11.70	32.34	43.92	36.09%
Neelum	7.23	35.91	33.09	35.92%

Q3. Concentration of Infrastructure Deficits

Share of affected schools by province

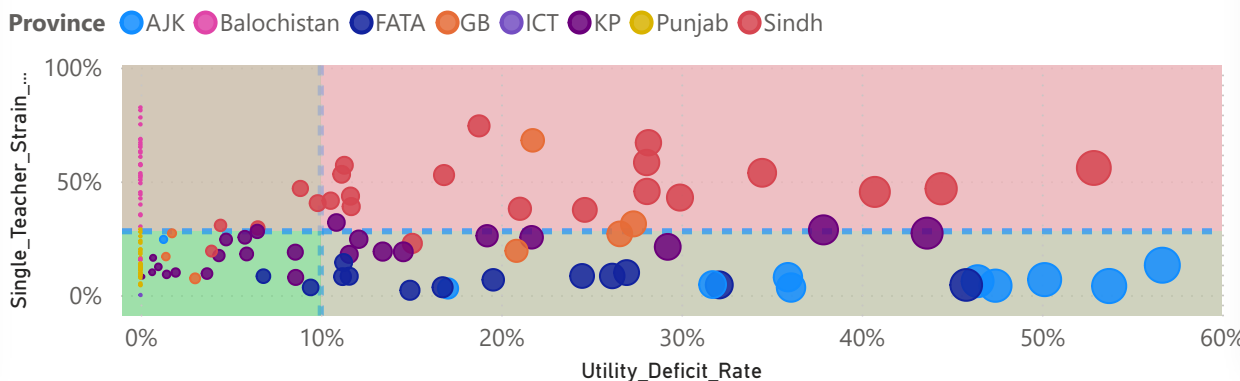


KEY INSIGHTS

Infrastructure deficits are concentrated in Tharparkar (Sindh), Kohistan (KP), and several districts in AJK & FATA. These areas need priority resource allocation.

Q4. Infrastructure Investment Priority

Districts with above-average infrastructure deficits and st...



KEY INSIGHTS

Districts in the red zone have both high utility deficit and high priority index, indicating urgent need for infrastructure investment.

Q4. Priority Districts for Infrastructure Investment

Higher coverage is better; higher No Facility indicates greater deficiency

Province	Utility_Deficit_Rate	Single_Teacher_Strain_Rate	Single_Classroom_Strain_Rate	Crisis_Schools_Volume	Infrastructure_Risk
AJK	46.64%	5.6%	15.9%	13.06	Critical
FATA	38.96%	4.4%	1.0%	3.12	High
KP	40.77%	28.4%	18.8%	3.26	High

KEY INSIGHTS

AJK and Sindh districts show very high teacher and classroom strain along with high utility deficit. These districts should be prioritized for infrastructure funding.



Government Action & Key Findings

Key Finding: Infrastructure deficits are concentrated across several districts in Sindh, with Tharparkar showing the highest affected-school volume. Haveli in AJK shows particularly severe facility coverage gaps, including a high proportion of schools with no facility.

Recommended Actions:

Prioritize infrastructure assessments and targeted facility investment in high-risk districts, especially where infrastructure deficits coincide with elevated single-teacher or single-classroom strain. Allocate resources based on both the severity of the deficit and the number of affected schools.



Executive Governance

Risk & Intervention

Infrastructure Audit

Infrastructure Analysis

Policy Impact & Socio-Economic

Socio Economic

Policy Impact & Socio-Economic Dashboard

Understanding Policy Context and Its Impact on Education Outcomes in Pakistan

Year

All

Province

All

Clear Filters

Education Budget-% GDP



2.52%

Global Terrorism Index



9.03

Drone Attacks



480

Average No Facility



10.26%

Average Education Score

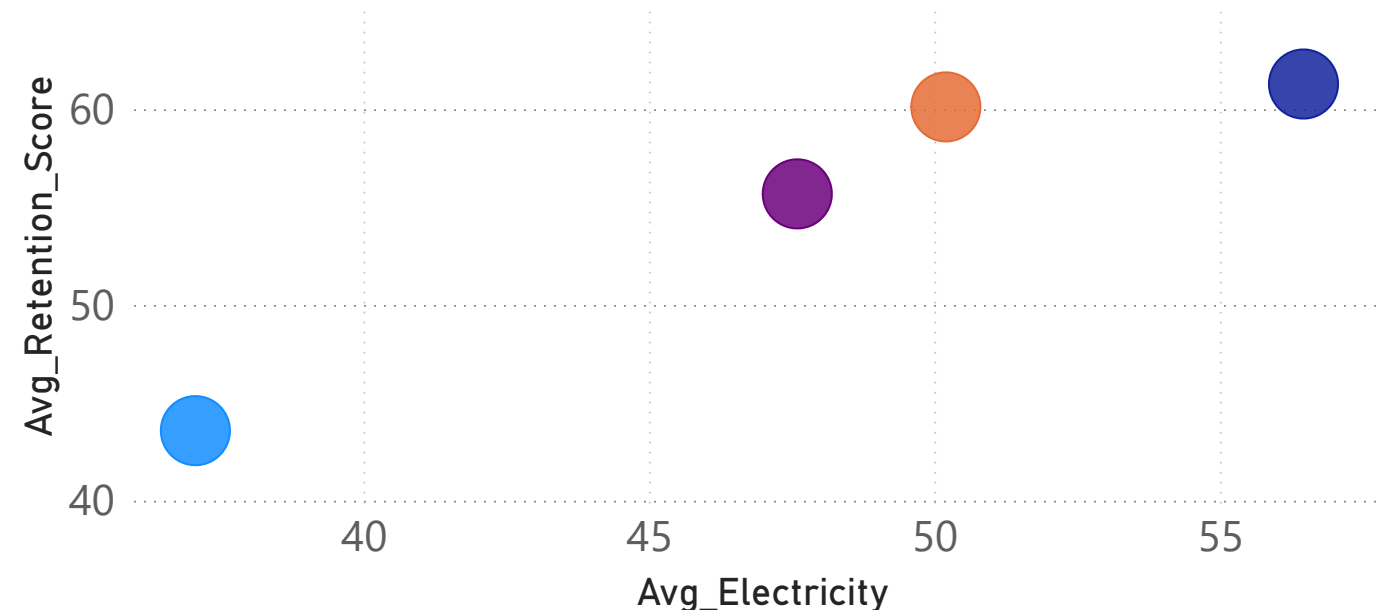


61.15

Facility Coverage vs Retention

Relationship between school electricity coverage and retention performance

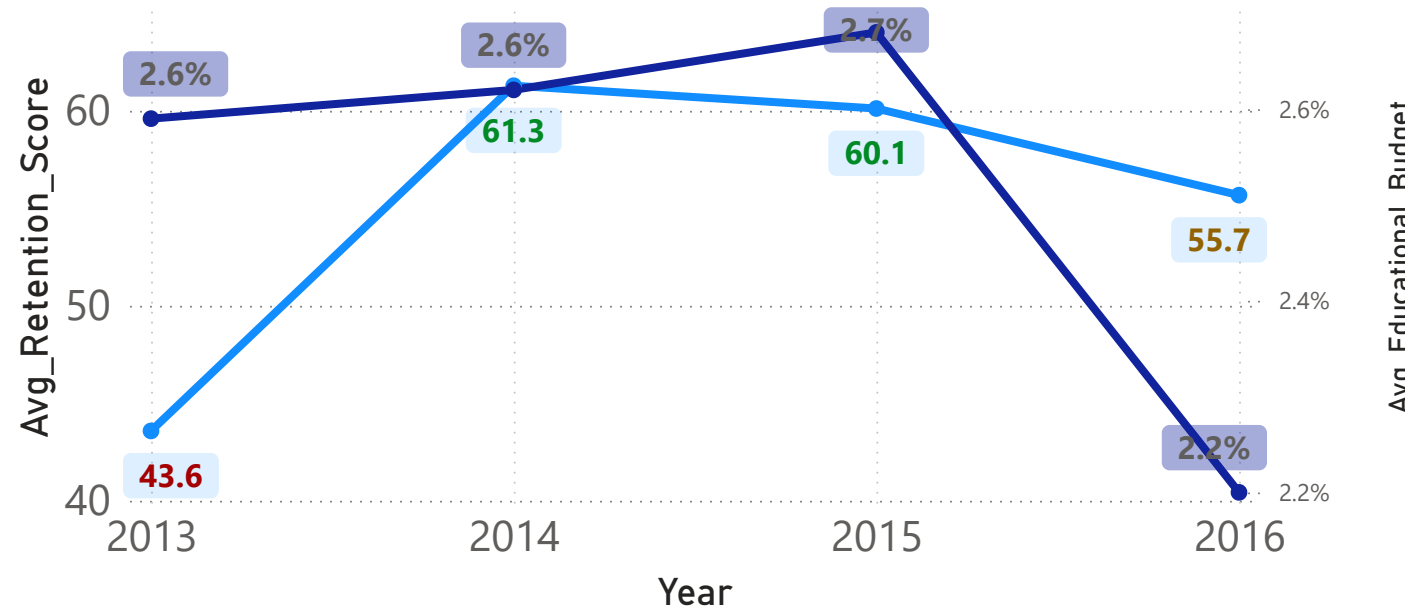
Year ● 2013 ● 2014 ● 2015 ● 2016



Education Investment vs Retention

Education budget allocation and student retention performance over time

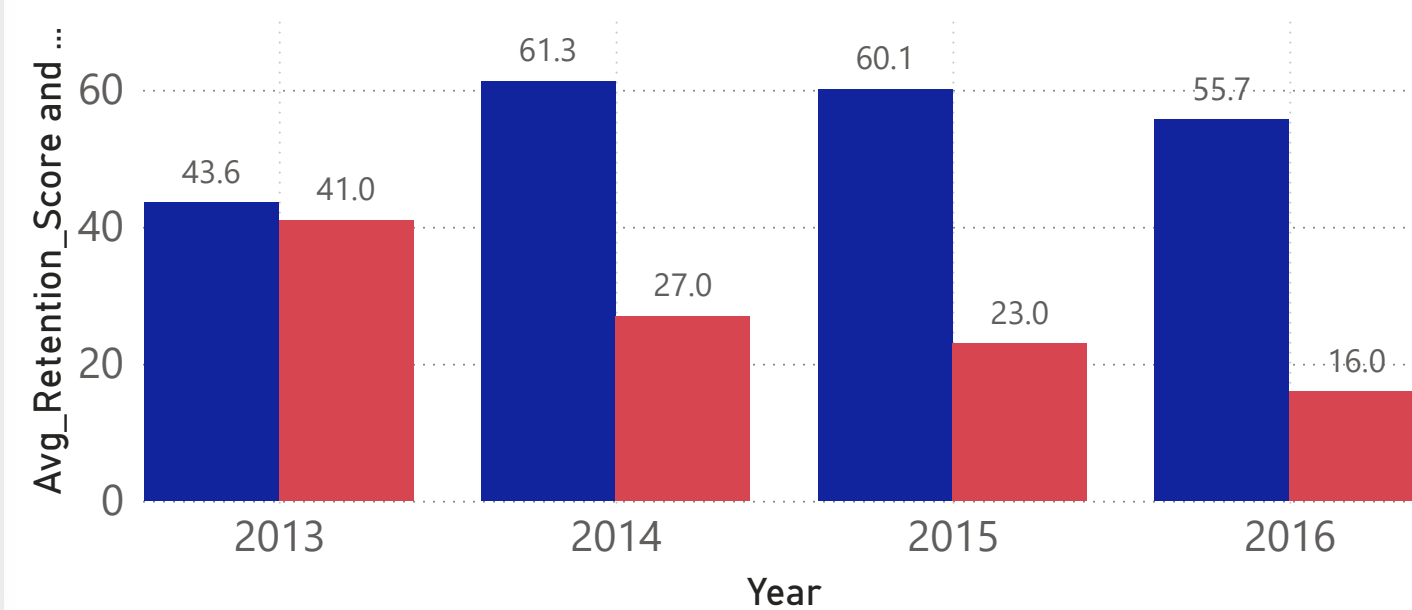
● Avg_Retention_Score ● Avg_Educational_Budget



Retention Score & Bomb Blast Trends

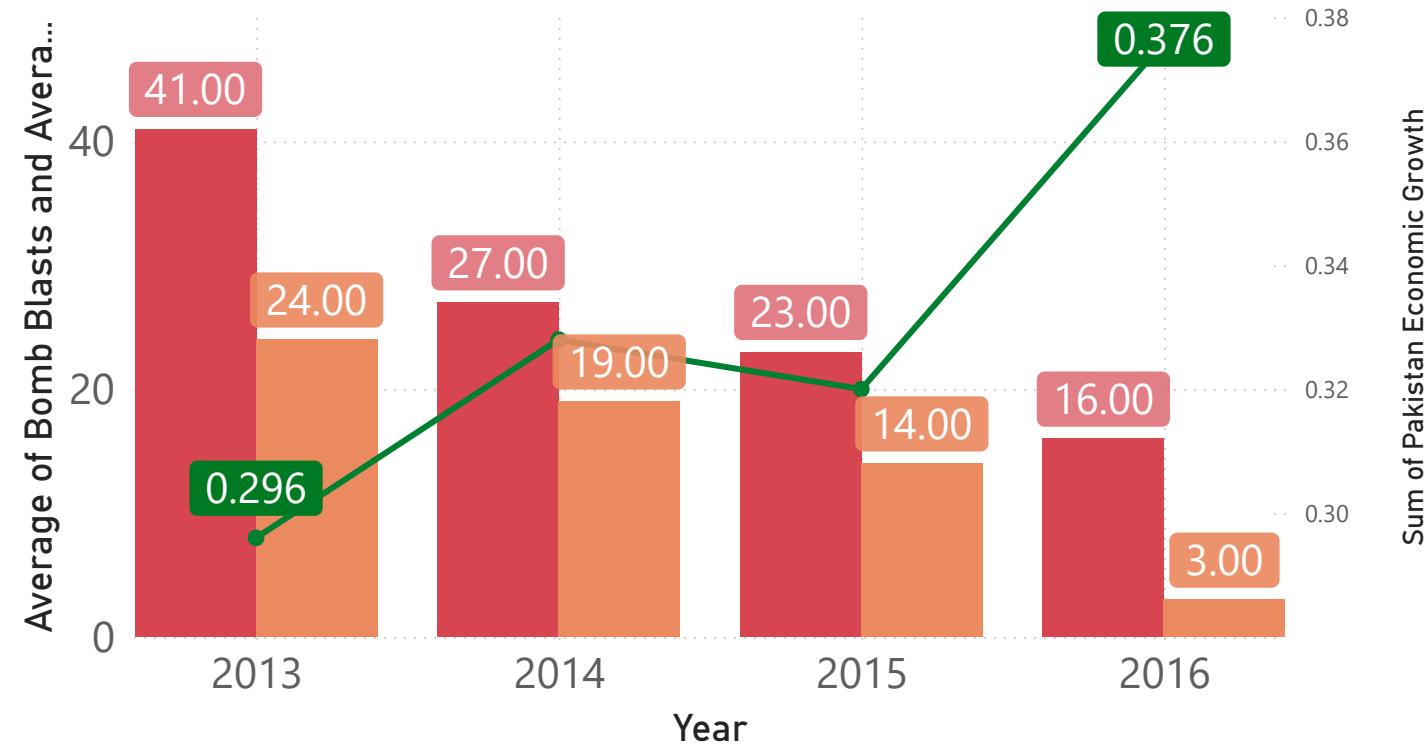
Comparing average retention performance with bomb blasts

● Avg_Retention_Score ● Average of Bomb Blasts



National Economic Growth vs National Terrorism

● Average of Bomb Blasts ● Average of Drone attacks ● Sum of Pakistan Economic Growth



Regional Infrastructure Policy Context

Comparing essential facility coverage and infrastructure gaps across provinces and districts

Province	Avg_No_Facility	Avg_Retention_Score	Avg_Learning_Score	Avg_Education_Score	Crisis_Schools_Volume
FATA	19.78%	38.8	37.3	47.50	10.29
Balochistan	0.00%	43.0	38.3	50.98	0.00
Sindh	20.57%	48.8	41.6	55.93	18.93
KP	10.73%	63.4	47.5	63.11	10.73
GB	14.68%	68.3	60.1	68.04	4.11
Punjab	0.00%	61.9	62.1	71.05	0.00
AJK	37.66%	71.1	60.5	74.00	15.06
ICT	0.00%	81.3	64.6	81.34	0.00
Total	10.26%	55.2	49.0	61.15	59.12

 Socio
Economic

Address single-teacher and single-classroom constraints.